Answers on
Booklet - 6

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

(a)(i)	horizontal distance = (4.0 x 20)		
	= 80 (m)		A1
(ii)	vertical fall: $s = ut + \frac{1}{2} at^2$		C1
	$= 0 + 0.5 \times 9.8 (4.0)^2$		C1
	= 78 (m)		A1
(iii)	horizontal component = 20 ($m\ s^{-1}$)		A1
(iv)	vertical component: $v = u + at$		
	$= 0 + 9.8 \times 4$		C1
	= 39 ($m\ s^{1}$)		A1
(b)(i)	friction/air resistance		
	lift		
	weight / force due to gravity	**any 2**	B2
(ii)	(gravitational) potential		B1
	converted to kinetic energy		B1
	and thermal energy / heat / work done against friction		B1
	two methods and two explanations		
	or four methods		
	e.g. increase speed down runway		
	larger surface area of skis		
	point skis upwards		
	lie as flat as possible / streamlining		B4

2.

(a)(i) energy due to position / height / above the ground **B1**
depends on gravitational field strength / weight **B1**
or mgh / wh B1 and symbols defined as mass, gravitational field strength and height / weight and height B1

(ii) energy **due to** movement / motion **B1**
depends on mass and speed **B1**
or $\frac{1}{2}mv^2$ B1 and symbols defined B1

(iii) work is the rate of doing work or rate of using energy **B1**
(work done/time taken)

(b)(i) pe = ke / mgh = $\frac{1}{2}$ mv^2 **C1**
$9.81 \times 130 = 0.5 \times v^2$ **C1**
$v = 50.5 \text{ m s}^{-1}$ (allow 50 or 51) **A1**
allow the use of $v^2 = u^2 + 2as$

(ii) mgh / t = 110000 **C1**
m/t =110000 / (9.81 x 130)
= 86.3 (kg s^{-1}) allow 86 **A1**
unit kg s^{-1} or kg **B1**

(iii) not all the pe of water will be converted to ke friction of water with ground or air resistance
not all the ke of water will be converted to ke of the wheel, the water retains some ke
friction in the rotation of the wheel not all ke converted to electrical energy
friction in the generator not all ke converted to electrical
any two **B2**

3.

(a)	force / load if proportional to extension	B1
(b)(i)	force constant = 100 / (40 x 10^{-3}) or equivalent	C1
	= 2500 N m^{-1} / kg s^{-2} (2.5 N mm^{-1})	A1
	unit penalty	-1
(ii)	work done = area under graph / (force x extension) / 2	C1
	= (120 x 48 x 10^{-3}) / 2	M1
	= 2.88 (2.9 to 2sf) (J)	A0
(c)(i)	k.e = ½ mv^2	C1
	v^2 = (2.9 x 2) / 0.015	C1
	v = 19.7 (ms^{-1}) (19.6 if 2.88 J is used)	A1
(ii)	(energy lost due to) friction in the gun	B1
	air resistance	B1
	(allow energy loss if type identified and place given)	
	(allow recoil of the gun)	

4.

a		(mv =) 300 ; kg m s^{-1} **or** N s	2
b	i	(The speed of the bar increases so) it is accelerated forwards/AW; this	1
		requires a resultant (forward) force/F = ma idea	1
	ii	(t = s/v = 3.0/0.60 =) 5.0 s	1
	iii	F = m (v – u)/t ; = 500 x 1.2/5.0 ; = 120 (N) *ecf b (iii)*	3

5.

a	**i**	Using F = ma gives P = 3ma hence a = P/3m	**1**
	ii	P/3	**1**
	iii 1	P/3m	**1**
	2	P/3	**1**
	iv 1	P/3	**1**
	2	2P/3	**1**
b	**i**	$mu = mv_1 + mv_2$	**1**
		$1/2mu^2 = 1/2mv_1^2 + 1/2mv_2^2$	**1**
	ii	some details of algebra/substitution to be shown resulting in e.g. $u = v_2$ and $u^2 = v_2^2$	**1**
c	**i**	all momentum is passed to block 2,(block 1 stops); then	**1**
		momentum is passed to block 3 (so block 2 does not move)/AW	**1**
		or argument in terms of k.e.	
	ii	block 1 bounces back; and blocks 2 & 3 move (to right/together)	**2**

6.

work done equals the <u>decrease</u> in GPE – <u>gain</u> in KE

distance = area under line
= (7.4 × 2.5) / 2 = 9.3 m (9.25 m)

or

acceleration from graph $a = 7.4/2.5$ (= 2.96)
and equation of motion $(7.4)^2 = 2 \times 2.96 \times s$ gives $s = 9.3$ (9.25) m

kinetic energy = $\frac{1}{2} mv^2$

$= \frac{1}{2} \times 75 \times (7.4)^2$

= 2100 J

potential energy = mgh
$h = 9.3 \sin 30°$
PE = 75 × 9.81 × 9.3 sin 30° = 3400 J

work done = energy loss
R = (3421 – 2054) / 9.3
= 150 (147) N

7.

add small mass to cause extension then remove mass to see if spring returns to original length

repeat for larger masses and note maximum mass for which, when load is removed, the spring does return to original length

Hooke's law requires force proportional to extension
graph shows a straight line, hence obeys Hooke's law

k = force / extension
= $(0.42 \times 9.81) / [(30 - 21.2) \times 10^{-2}]$
= 47 (46.8) Nm^{-1}

8.

GPE: energy of a mass due to its position in a gravitational field
KE: energy (a mass has) due to its motion / speed / velocity

9.

E = stress / strain
$E = 0.17 \times 10^{12}$
stress $= 0.17 \times 10^{12} \times 0.095 / 100$
$= 1.6(2) \times 10^{8}$ Pa

force = (stress × area) = $1.615 \times 10^{8} \times 0.18 \times 10^{-6}$
= 29(.1) N

10.

weight = 452 × 9.81
component down the slope = 452 × 9.81 × sin 14°
= 1072.7 = 1070 N

$F = ma$
$T - (1070 + 525) = 452 \times 0.13$
$T = 1650\ (1653.76)$ N *any forces missing 1/3*

1. $s = ut + \frac{1}{2}at^2$ hence $10 = 0 + \frac{1}{2} \times 0.13t^2$
$t = [(2 \times 10) / 0.13]^{1/2} = 12.4$ or 12 s

2. $v = (0 + 2 \times 0.13 \times 10)^{1/2} = 1.61$ or $1.6\ \mathrm{m\,s^{-1}}$

11.

triangle drawn with correct shape (incorrect arrows loses this mark)
$T_1 = 5.4 \pm 0.2$ N
$T_2 = 4.0 \pm 0.2$ N